

# Applications of AI in Africa (*Hargricon: A Digital Platform Connecting Farmers and Buyers*): A Proposal

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## I. INTRODUCTION

The Rwandan economy relies heavily on agriculture that makes up more than 40% of the labor force and also makes important contributions to economic development and food security [1]. However, smallholder farmers in the country experience limited access to markets, inefficient supply chains, inadequate storage, and lack of market information.

Approximately 3 million tons of food are wasted every year in Rwanda, representing 40% of the total food production and 12% of the yearly GDP, whereas 19% of households are still suffering from food insecurity [2]. Thus, there is a threat of food insecurity, farmer's profitability, and agricultural productivity.

Data-driven decisions through AI can provide a way to circumvent the challenges outlined above. Digital markets that have AI can connect farmers and consumers directly while also providing forecasts about price trends, market demand, and harvesting.

## II. PROBLEM STATEMENT

Smallholder farmers in Rwanda still find it difficult to market their agricultural produce. Most of the smallholder farmers take their products to the markets without assured buyers, and this leads to increased costs, increased selling time, lack of bargaining power, and spoilage of the perishable commodities. [2].

These losses are quite substantial as maize and rice losses are 25% and 18% respectively while tomato losses are as high as 49% [2]. Post-harvest losses for perishable products in Sub-Saharan Africa are as high as 30-40% and poor market coordination [3].

Lack of availability of market information in time also adds to the problem. As per Rwanda's Ministry of Agriculture and Animal Resources (MIN-AGRI), poor post-harvest management systems and lack of market information play a vital role in the inflating prices and lower income of smallholder farmers. [4].

## III. PROPOSED SOLUTION

Hargricon is an AI-powered agricultural marketplace connecting smallholder farmers with buyers through a digital platform. The AI system analyzes market prices, seasonal trends, crop availability, and location data to provide intelligent price recommendations and demand forecasts, enabling better decisions, reducing food waste, and improving supply chain efficiency.

## IV. AIM AND OBJECTIVES

**Aim:** To create a digital platform driven by AI that increases smallholder farmers' access to markets and lowers post-harvest losses in Rwanda.

**Objectives:**

- Build a web-app that connects farmers directly with buyers.

- Integrate an AI system that predicts market demand and recommends competitive pricing.
- Improve agricultural market efficiency by reducing food spoilage and shortening the time between harvest and sale.

## V. METHODOLOGY

Hargricon will go through development via the requirements gathering, designing, implementation, testing, and evaluation process. For the web interface and backend of the application, Next.js will be used; as for the database for storing data about users and products, PostgreSQL (Neon) will be utilized, as well as Neon Auth for authentication. The app will be deployed on Vercel.

For the AI part, current prices in agriculture will be obtained from sites like e-Soko and other marketplaces and compared for crops and location, and selling prices will be suggested for farmers.

### A. List of Assumptions

- Internet access is available to the farmers.
- Users will give accurate and up-to-date information regarding produce that is available for sale.
- Past market information will be adequate to train the initial predictive model.
- The buyers will use digital means for sourcing agricultural products.

### B. List of Resources

Software:

- Frontend: NextJS
- Backend: NextJS API Route
- Database: PostgreSQL (via Neon)
- Authentication: NeonAuth
- Deployment: Vercel
- AI model: Gemini AI Design Inspo: Dribbble

Data Sources:

- Esoko agricultural price trends platform
- Public agricultural market information sources

TABLE I: Member Participation and Tasks

| No. | Milestone                  | Responsible          |
|-----|----------------------------|----------------------|
| 1.  | Web Development            | Lateef Quadri        |
| 2.  | Team Management and Design | Abdullahi Abdulkadir |
| 3.  | Research and Writing       | Adarijo Rasheed      |
| 4.  | Database                   | Uwayo Olga           |
| 5.  | Writing and Editing        | Kayiranga Sandrine   |
| 6.  | Resource Coordination      | Girihirwe Anaella    |

## VI. WORK PLAN

The different sections of how the work was split are defined in Table I.

## VII. AI USE DISCLOSURE

Gemini AI will function as a market intelligence tool at Hargricon by collecting agricultural price information from Esoko and other sources to assist farmers and buyers in making good decisions. Also, ChatGPT has been done to support brainstorming and research. All technical content, project design decisions, reference verifications, and final reviews were performed by the project team

## REFERENCES

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